

2023 Ph H1 Q11

Section: Particles and Waves

Topic: The Standard Model

Brief summary of the question

Identify the type of particle from its quark composition and charge.

Worked solution

- Quark composition: up, up, down (u, u, d).
- This is three quarks, so it is a baryon (baryons are hadrons made of three quarks).
- Charge calculation: up ($+2/3 e$) + up ($+2/3 e$) + down ($-1/3 e$) = $+1 e$.
- A baryon with charge $+1 e$ is a proton.

Final answer

C — baryon.

Revision tips

Baryons (three quarks) include protons (uud) and neutrons (udd).

Charge comes from adding quark charges.

Protons are stable; neutrons decay outside nuclei.